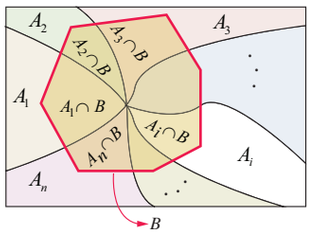
12.8 Bayes' Theorem



1702-1761

Thomas Bayes was an English statistician, philosopher and Presbyterian minister who is known for formulating a specific case of a theorem. Bayesian methods stem from the principle of linking prior (before conducting experiment) probability and conditional probability (likelihood) to posterior (after conducting experiment) probability via Bayes' rule. Bayesian probability is the name given to several related interpretations of probability as an amount of epistemic confidence – the strength of beliefs, hypotheses etc., rather than a frequency.



Theorem 12.11 (Bayes' Theorem)

If $A_1, A_2, A_3, ..., A_n$ are mutually exclusive and exhaustive events such that $P(Ai) > 0, \quad i = 1,2,3,\ldots.n$ and B is any event in which $P(B) > 0$, then

$$P(A_i / B) = \frac{P(A_i)\, P(B / A_i)}{P(A_1)\, P(B / A_1) + P(A_2)\, P(B / A_2) + \cdots + P(A_n)\, P(B / A_n)}$$

Proof

By the law of total probability of B we have

$$P(B) = P(A_1) \cdot P(B / A_1) + P(A_2) \cdot P(B / A_2) + \cdots + P(A_n) \cdot P(B / A_n)$$

and by multiplication theorem $P(A_i \cap B) = P(B/A_i) P(A_i)$

By the definition of conditional probability,

$$P(A_i / B) = \frac{P(A_i \cap B)}{P(B)}$$

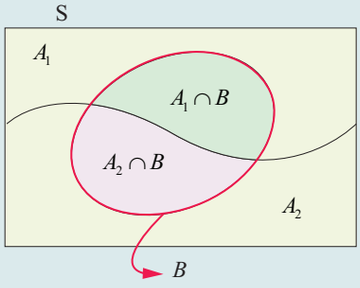
$$P(A_i / B) = \frac{P(B/A_i) P(A_i)}{P(A_1) \cdot P(B / A_1) + P(A_2) \cdot P(B / A_2) + \cdots + P(A_n) \cdot P(B / A_n)} \quad \text{(using formulae)}$$

The above formula gives the relationship between $P(A_i / B)$ and $P(B / A_i)$

Example 12.26

A factory has two machines I and II. Machine I produces 40% of items of the output and Machine II produces 60% of the items. Further 4% of items produced by Machine I are defective and 5% produced by Machine II are defective. An item is drawn at random. If the drawn item is defective, find the probability that it was produced by Machine II. (See the previous example, compare the questions).

Solution



Let A_1 be the event that the items are produced by Machine-I, A_2 be the event that items are produced by Machine-II. Let B be the event of drawing a defective item. Now we are asked to find the conditional probability $P(A_2/B)$. Since A_1, A_2 are mutually exclusive and exhaustive events, by Bayes' theorem,

$$P(A_2 / B) = \frac{P(A_2)\, P(B / A_2)}{P(A_1)\, P(B / A_1) + P(A_2)\, P(B / A_2)}$$

12_11th-Maths-Vol2-Unit12.indd 261 16-12-2022 11:11:50